

Vector DB: (Search) Technical Specification

Version 3.4 – Decision-Oriented, Hardware-Aware, Evolutionary Narrative

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0. Notation Table

Symbol	Meaning
N	Number of stored vectors
D	Vector dimensionality
B	Bytes per element
n	Number of vectors touched (algorithmic skipping)
n_probe	Number of vectors probed in IVF cluster
D_eff	SIMD-aligned effective dimension
B_quant	Bytes per element after quantization
Γ	Locality multiplier (hardware)
Λ	Software / glue code overhead
Parallelism	Number of CPU cores / threads
Φ	Unified system strain
$\Phi_{\text{distributed}}$	Distributed system strain
InstrCost	CPU instruction cost per operation
K	Number of top results requested
S	Number of shards / nodes

1. Historical Foundation: Unit Cost of Search

Every vector search reduces to **CPU ↔ Memory transactions**.

Computational Cost:

$$C_{\text{Search}} = N \cdot D \cdot \text{MathCost}$$

Memory Transfer Cost:

$$M_{\text{Total}} = N \cdot D \cdot B + \text{Metadata}$$

Where:

- N = number of stored vectors
- D = vector dimensionality
- B = bytes per element
- MathCost = cost per floating point / integer / bitwise operation

Observation:

- Naive linear search (K-NN) is $O(N)$. As datasets grow, the CPU stalls waiting for RAM — the **Memory Wall**.
 - This forms the baseline; all later equations refine these fundamental costs.
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2. Evolutionary Derivation: From Unit Cost to Unified System Strain

2.1 Naive Linear Search (Brute Force)

Unit operation: compare **one query vector to one stored vector**:

$$C_{\text{Unit}} = D \cdot \text{MathCost} + D \cdot B \cdot \text{MemoryAccessCost}$$

Total cost for N vectors:

$$C_{\text{Total}} = N \cdot C_{\text{Unit}}$$

Observation: Linear in N , dominated by memory for large datasets $\rightarrow$ Memory Wall.

2.2 Clustered Indexing (IVF)

Partition vectors into k clusters. Search involves:

1. Find nearest cluster centroid: $k \cdot C_{\text{Unit}}$
2. Search within selected clusters: $n_{\text{probe}} \cdot N/k \cdot C_{\text{Unit}}$

$$C_{\text{Total}} = k \cdot C_{\text{Unit}} + n_{\text{probe}} \cdot N/k \cdot C_{\text{Unit}}$$

Effect: Reduces effective $N \rightarrow n$, but adds centroid search overhead.

2.3 Graph-Based Search (HNSW)

Search cost becomes **logarithmic** along graph paths:

$$C_{\text{Total}} \approx \log(N) \cdot C_{\text{Unit}}$$

Effect: Visits only relevant neighbors → massive reduction in computations.

2.4 Algorithmic Skipping & Clustering

- Introduce **skipping factor (n)**: $n \ll N$
- Partition vectors into clusters (IVF / HNSW)
- Search only relevant neighborhoods

Efficiency Ratio (ER):

$$ER = \frac{\text{Data Used}}{\text{Data Moved}}$$

Result: Search cost reduced from $O(N) \rightarrow O(\sqrt{N})$ or $O(\log N)$.

Note: First major lever to reduce system strain without changing hardware.

2.5 Representation Evolution: Quantization

Method	Bytes/Dimension	Effect
Scalar Quantization (SQ)	1	4× memory savings, minor accuracy loss
Product Quantization (PQ)	0.02–0.1	50× memory savings, fits billion-scale datasets
Binary Quantization (BQ)	0.125	Ultra-fast XOR math, high-speed filtering

- **SQ:** reduces B
 - **PQ:** splits $D \rightarrow D_{\text{sub}}$, reduces memory footprint
 - **BQ:** compresses both B and D, speeds computation
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3. Hardware & Software Awareness: Locality (Γ) and Glue Code (Λ)

Vector search cost is **not only algorithmic**; memory access patterns and software overhead can dominate latency, especially for small or distributed datasets.

We capture these effects with two factors:

1. **Locality Multiplier (Γ)** — models cache/memory efficiency
2. **Glue Code Overhead (Λ)** — models software and network delays

3.1 Locality Multiplier (Γ)

Memory Access Pattern	Γ Value	Effect / Comment
Sequential / Flat Scan	1.0	Prefetching reduces cache misses; memory-efficient
Clustered (IVF)	2–5	Mostly sequential within clusters; some cache misses
Random / Graph (HNSW)	5–10	Pointer-chasing causes frequent cache misses; CPU stalls

Notes:

- Γ is **empirical**, depends on CPU, RAM speed, cache size
- Converts algorithmic cost into **hardware-realistic latency**
- Higher $\Gamma \rightarrow$ more CPU cycles wasted per vector touched

3.2 Glue Code & Software Overhead (Λ)

Overhead Source	Typical Latency per Vector	Comment
Serialization / Deserialization	1–10 μ s	Converts data between application layers
Context Switching (OS / Thread)	5–50 μ s	OS scheduling, task switching
Network Round-Trip (Local / Cluster)	50–500 μ s	Depends on cluster topology and network

Notes:

- Λ dominates small datasets ($N < 10k$), making flat in-process scans faster than distributed DB
- Total software overhead scales linearly with vectors touched

3.3 Unified Cost Contribution

$$\Phi = \Lambda + \Gamma \cdot \frac{n \cdot D_{\text{eff}} \cdot B_{\text{quant}} \cdot \text{InstrCost}}{\text{Parallelism}}$$

Where:

- n = vectors actually touched (skipping applied)
- D_{eff} = SIMD-aligned dimensions
- B_{quant} = bytes per element after quantization
- InstrCost = CPU instruction cost (Bitwise > Integer > Float)
- Parallelism = number of CPU cores / threads

Interpretation:

- Λ handles **software latency**
 - Γ handles **hardware memory penalties**
 - Fraction captures **parallelized vector computation**
 - Φ gives a **decision-oriented metric** for deployment
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4. Extending to Clusters / Multi-Node Systems

$$\Phi_{\text{dist}} = \max(\Phi_{\text{shard}}) + \text{Cost}_{\text{merge}} + \Lambda_{\text{network}}$$

- Merge cost: $O(K \log S)$
 - S = number of shards
 - Merge = distributed Top-K aggregation
 - Distribution tax appears once $N \gtrsim 100M$
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4.1 Multi-Node Equation ($\Phi_{\text{Distributed}}$)

$$\Phi_{\text{Distributed}} = \Phi_{\text{Local}}/S + (\Lambda_{\text{Network}} + \text{Cost}_{\text{Merge}}(K, S))$$

- Λ_{Network} = network latency
- $\text{Cost}_{\text{Merge}}$ = time to collect Top-K from shards

4.2 Scaling Simulation (100 Million Vectors)

Nodes (S)	RAM per Node	Latency (Local)	Latency (Merge)	Total Latency
1	400 GB	500 ms	0 ms	500 ms
10	40 GB	50 ms	5 ms	55 ms
50	8 GB	10 ms	15 ms	25 ms

Observation:

- Increasing S reduces per-node search time linearly
- Merge cost grows with S; coordinator becomes bottleneck
- Adding nodes beyond threshold can **increase total latency**

Note: These values are simulated / example estimates.

5. Practical Decision Matrix

Scale (N)	Primary Bottleneck	Recommended Strategy	Representation (B / D)
Small (<10k)	Glue Code (Λ)	Flat / In-Process	float32 (no compression)
Mid (<1M)	Cache Locality (Γ)	IVF / Flat-GPU	int8 (Scalar Quantization)
Large (>10M)	Search Space (n)	HNSW (Graph)	PQ (Product Quantization)
Massive (>100M)	RAM / Network Budget	Disk-Based IVF + Sharding	BQ (Binary Quantization)

6. Verification Checklist

- SIMD alignment: D aligned to 16
 - Bandwidth: $\text{Data Size} / \text{Target Latency} \leq \text{RAM Bandwidth}$
 - Locality: Index size vs L3 Cache
 - Glue: Λ not dominating small N searches
 - Quantization: B_{quant} matches memory/accuracy targets
 - Parallelism: sufficient cores for n vectors processed
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7. Cost Modeling

$$\text{Monthly_Cost} = N \cdot D_{\text{eff}} \cdot B_{\text{quant}} \cdot \text{GB_Rate}$$

- Memory dominates cost $\rightarrow$ prioritize reducing B_{quant}

8. Decision Flow

Input: N , D , B , Target Latency

- ↓ Compute D_{eff} and size
- ↓ Compute $\Phi_{\text{min}} = \text{Size} / \text{Bandwidth}$
- ↓ Check $N < 15k$? $\rightarrow$ Flat / In-Process
- ↓ Select index based on α (Recall)
- ↓ Adjust n , B_{quant} , Γ , Parallelism
- ↓ Compute Φ and verify $\leq$ Target
- ↓ Deploy Locally or Distributed ($\Phi_{\text{distributed}}$)

9. Evolutionary Summary

- Unit Cost: Compute + memory $\rightarrow$ Memory Wall
- Algorithmic Skipping: Reduce effective N
- Quantization: Reduce $D \times B$ memory footprint
- Hardware Awareness: Locality multiplier Γ
- Software Awareness: Glue tax Λ
- Unified System Strain: Φ combines all variables
- Cluster / Shard Extension: $\Phi_{\text{distributed}}$ for distributed setups

10. Notes / Recommendations

- Always simulate Φ before deployment to choose shard count and quantization levels
- For datasets $> 10M$ vectors, PQ or BQ is essential
- Keep Λ under control for small N ; avoid unnecessary network/distributed layers
- Hardware-specific tuning (SIMD, cache sizes, RAM bandwidth) will directly impact Φ
- Document decisions per project: N , D , B , α targets, resulting Φ

11. Vector Search: Evolution Math Table

Stage	Unit / Operation	Total Cost Equation	Key Observation / Evolutionary Insight
1. Naive Linear Search	Compare 1 query vector to 1 stored vector	$C_Unit = D \cdot \text{MathCost} + D \cdot B \cdot \text{MemoryAccessCost}$ $C_Total = N \cdot C_Unit$	Linear in N; dominated by memory → Memory Wall
2. Clustered Indexing (IVF)	Search within selected clusters	$C_Total = k \cdot C_Unit + n_probe \cdot N/k \cdot C_Unit$	Reduces effective $N \rightarrow n$; adds centroid overhead; first major “skip” lever
3. Graph-Based Search (HNSW)	Traverse neighbors along graph paths	$C_Total \approx \log(N) \cdot C_Unit$	Logarithmic cost; visits only relevant neighbors; massive reduction
4. Quantization (SQ / PQ / BQ)	Reduce memory footprint per vector	$C_PQ_Unit = D_sub \cdot B_sub \cdot \text{MathCost} + \text{LookupTableOverhead}$	SQ reduces B, PQ splits $D \rightarrow D_sub$, BQ compresses both; speeds computation
5. Locality & Software Awareness	Hardware / software adjustments	$\Gamma \ \& \ \Lambda$	Converts algorithmic cost into hardware-realistic cost; small N dominated by Λ
6. Unified System Strain (Φ)	Combines all factors	$\Phi = \Lambda + \Gamma \cdot (n \cdot D_eff \cdot B_quant \cdot \text{InstrCost}) / \text{Parallelism}$	Captures algorithmic skipping, quantization, locality, software overhead, parallelism; decision-oriented metric

12. Vector Search Decision Table

Scale (N)	Primary Bottleneck	Recommended Algorithm / Index	Memory & Compute (B / D)	Γ	Λ	Notes / Deployment Guidance
Small (<10k)	Software / Glue Code	Flat / In-Process	float32 (no compression)	1.0 (sequential)	Dominant	Flat scan faster than distributed DB; minimize network layers
Medium (10k–1M)	Cache Locality	IVF / Flat-GPU	int8 (Scalar Quantization)	2–5 (clustered)	Moderate	Partition into clusters; sequential access within clusters improves Γ
Large (1M–10M)	Search Space (n)	HNSW (Graph)	PQ (Product Quantization)	5–10 (random)	Minor	Algorithmic skipping ($n \ll N$) critical; reduces

Scale (N)	Primary Bottleneck	Recommended Algorithm / Index	Memory & Compute (B / D)	Γ	Λ	Notes / Deployment Guidance
Massive (>10M–100M)	RAM / Network Budget	Disk-Based IVF + Sharding	BQ (Binary Quantization)	5–10 (random)	Significant	effective vectors touched Distributed nodes required; merge cost and network latency must be monitored
Ultra-Massive (>100M)	RAM & Network Limitations	Sharded / Multi-Node HNSW + PQ/BQ	PQ/BQ	5–10 (random)	High	Plan Φ _distributed carefully; consider hierarchical merging & local caching

12.1 Explanation of Columns

- **Scale (N):** Number of vectors
- **Primary Bottleneck:** Dominant factor limiting performance
- **Recommended Algorithm / Index:** Best practice based on scale
- **Memory & Compute (B / D):** Quantization / representation
- **Γ :** Hardware-adjusted factor for memory access
- **Λ :** Software overhead / glue code
- **Notes / Deployment Guidance:** Practical tuning guidance

12.2 How to Use the Table

1. Determine dataset size (N) and target latency
2. Identify primary bottleneck
3. Pick algorithm & representation
4. Apply hardware-aware tuning: reduce Γ with sequential access or cache-aware clustering
5. Reduce Λ by minimizing serialization / network hops
6. Compute Φ or Φ _distributed to verify latency targets
7. Iterate quantization, parallelism, cluster/shard count to balance accuracy, memory, and speed

13. Minor caveats / potential improvements:

- **Γ values (5–10 for HNSW):** These are rough empirical estimates; actual cache misses can vary significantly depending on CPU, RAM bandwidth, and vector size. You might want a **range and profiling guidance** rather than a single number.
 - **Λ dominance for $N < 10k$:** True for interpreted languages like Python or networked setups, but less significant in fully optimized C++ or GPU in-process searches. Worth noting as an assumption.
 - **Distributed merge cost:** $O(K \log S)$ is for naive Top-K merging. Some systems can do partial pre-aggregation to reduce merge cost — you could note this optional optimization.
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14. Things to be cautious about in practice:

1. **Parameter tuning:**
 - n_probe , cluster count (k), graph connectivity ($efConstruction$ / $efSearch$ for HNSW), and quantization levels will significantly affect Φ .
 - These are not fully formulaically derived — simulation/profiling is still required.
2. **Edge cases:**
 - Very small vectors (e.g., $D < 16$) might make SIMD alignment less meaningful.
 - GPU / mixed CPU-GPU setups are not explicitly modeled. Φ assumes CPU-like latency.
3. **Network variability:**
 - $\Lambda_{Network}$ is given as 50–500 μs ; actual latency in cloud deployments can fluctuate by an order of magnitude. Consider including **confidence intervals** or **percentile latency** for production planning.
4. **Memory bandwidth:**
 - Current model assumes memory bandwidth as a soft cap in Φ ; in reality, **saturating RAM channels** can create nonlinear latency effects (especially in multi-threaded or NUMA setups).